Week 10 Handout: Submodular Flows and Applications

1 Motivation

We have seen several consequences of submodularity in the study of polymatroids and polymatroid intersection. In this lecture, we discuss another place where submodularity appears naturally: network flows. The main object is a flow-like vector whose net flow across every relevant cut is controlled by a crossing submodular function.

The reason this model is useful is that it is both expressive and algorithmically well behaved. It contains ordinary circulations, gives another proof of the integrality of polymatroid intersection, and also yields clean proofs of classical min-max theorems in directed graphs.

Throughout the handout, let $D = (V, A)$ be a directed graph. For $U \subseteq V$, write

$$\delta^-(U) = \{a = xy \in A : x \notin U,\ y \in U\}, \qquad \delta^+(U) = \{a = xy \in A : x \in U,\ y \notin U\}.$$

For a vector $x \in \mathbb{R}^A$ and a set of arcs $F \subseteq A$, we use the convention

$$x(F) \coloneqq \sum_{a \in F} x(a).$$

2 Crossing Submodularity and Submodular Flows

Definition 2.1 (Crossing family)**.** A family $\mathcal{C} \subseteq 2^V$ is called a *crossing family* if for every $X, Y \in \mathcal{C}$ satisfying

$$X \cap Y \neq \emptyset \qquad \text{and} \qquad X \cup Y \neq V,$$

we also have

$$X \cap Y \in \mathcal{C} \qquad \text{and} \qquad X \cup Y \in \mathcal{C}.$$

The point is that we only require closure under intersection and union for pairs that genuinely cross. We do not impose full lattice closure on all pairs of sets.

Definition 2.2 (Crossing submodular function)**.** Let $\mathcal{C} \subseteq 2^V$ be a crossing family. A function $f : \mathcal{C} \to \mathbb{R}$ is *crossing submodular* on $\mathcal{C}$ if for every $X, Y \in \mathcal{C}$ satisfying

$$X \cap Y \neq \emptyset \qquad \text{and} \qquad X \cup Y \neq V,$$

we have

$$f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y).$$

Definition 2.3 (Submodular flow feasibility system)**.** Let $D = (V, A)$ be a digraph, let $\mathcal{C} \subseteq 2^V$ be a crossing family, and let $f : \mathcal{C} \to \mathbb{R}$ be crossing submodular. The associated *submodular flow system* is

$$x(\delta^-(U)) - x(\delta^+(U)) \leq f(U) \qquad \forall U \in \mathcal{C}.$$

A vector $x \in \mathbb{R}^A$ satisfying these inequalities is called feasible for the submodular flow system. In applications we often also impose box constraints

$$\ell(a) \le x(a) \le r(a) \qquad \forall a \in A.$$

Example 2.4 (Why this is a flow model)**.** The expression

$$x(\delta^-(U)) - x(\delta^+(U))$$

is the net amount of flow entering U. Thus a submodular flow system controls the net imbalance across a family of cuts. Different choices of $\mathcal{C}$ and f encode different global constraints.

- Ordinary circulation constraints are obtained by controlling the imbalance at each single vertex.
- Polymatroid intersection constraints can be represented by taking two copies of the ground set and using arcs between the two copies.
- Directed cut covering and packing problems can be written using negative right-hand sides and a change of variables.

3 A Recap on TDI Systems

We next recall the integrality framework used in the Edmonds-Giles theorem.

Definition 3.1 (Totally dual integral system)**.** A rational system of inequalities $Bx \le b$ is *totally dual integral*, or TDI, if for every integral vector c such that the primal optimum

$$\max\{c^T x : Bx \le b\}$$

exists and is finite, the dual linear program

$$\min\{yb : y \ge 0,\ yB = c\}$$

has an integral optimum y^*.

Definition 3.2 (Box-TDI system)**.** A system $Bx \le b$ is *box-TDI* if the augmented system

$$Bx \le b, \qquad \ell \le x \le r$$

is TDI for every rational choice of lower and upper bounds ℓ, r.

Theorem 3.3 (TDI implies integrality)**.** *If $Bx \le b$ is TDI and b is integral, then the polyhedron*

$$\{x : Bx \le b\}$$

is integral.

This is one of the main reasons TDI systems are useful: they certify integrality by proving integrality of the dual, rather than by explicitly describing vertices of the primal polyhedron.

4 The Edmonds-Giles Theorem

Theorem 4.1 (Edmonds-Giles)**.** *Let $D = (V, A)$ be a digraph, let $\mathcal{C} \subseteq 2^V$ be a crossing family, and let $f : \mathcal{C} \to \mathbb{R}$ be crossing submodular. Then the submodular flow system*

$$x(\delta^-(U)) - x(\delta^+(U)) \le f(U) \qquad \forall U \in \mathcal{C}$$

is box-TDI.

We give only the proof sketch, since the complete proof requires some work with network matrices.

Let M be the cut-incidence matrix whose rows are indexed by sets $U \in \mathcal{C}$ and columns are indexed by arcs $a \in A$, with

$$M(U,a) = \begin{cases} 1, & a \in \delta^-(U), \\ -1, & a \in \delta^+(U), \\ 0, & \text{otherwise.} \end{cases}$$

Then the submodular flow constraints are simply

$$Mx \le f.$$

Adding box constraints gives

$$\begin{pmatrix} M \\ I \\ -I \end{pmatrix} x \le \begin{pmatrix} f \\ r \\ -\ell \end{pmatrix}.$$

For an integral weight vector $w \in \mathbb{Z}^A$, consider the primal linear program

$$\max \left\{ w^T x : Mx \le f, \ell \le x \le r \right\}.$$

Its dual is

$$\begin{aligned} \min \quad & \sum_{U \in \mathcal{C}} y(U) f(U) + \sum_{a \in A} p(a) r(a) - \sum_{a \in A} q(a) \ell(a) \\ \text{s.t.} \quad & \sum_{U : a \in \delta^-(U)} y(U) - \sum_{U : a \in \delta^+(U)} y(U) + p(a) - q(a) = w(a) \quad \forall a \in A, \\ & y(U), p(a), q(a) \ge 0 \qquad\qquad \forall U \in \mathcal{C},\ a \in A. \end{aligned}$$

To prove box-TDI, it is enough to show that whenever this dual has an optimum, it has an integral optimum.

Uncrossing the support of the dual

A set family $\mathcal{F} \subseteq 2^V$ is called *cross-free* if for every $X, Y \in \mathcal{F}$, at least one of the following holds:

$$X \subseteq Y, \qquad Y \subseteq X, \qquad X \cap Y = \emptyset, \qquad X \cup Y = V.$$

Claim 4.2 (Uncrossing)**.** *There is an optimal dual solution (y, p, q) such that*

$$\operatorname{supp}(y) = \{U \in \mathcal{C} : y(U) > 0\}$$

is cross-free.

Proof sketch. Suppose $X, Y \in \operatorname{supp}(y)$ cross, meaning

$$X \cap Y \neq \emptyset \qquad \text{and} \qquad X \cup Y \neq V.$$

Since $\mathcal{C}$ is a crossing family, both $X \cap Y$ and $X \cup Y$ belong to $\mathcal{C}$. Choose

$$0 < \varepsilon \leq \min\{y(X), y(Y)\},$$

and modify y by subtracting ε from $y(X)$ and $y(Y)$, while adding ε to $y(X \cap Y)$ and $y(X \cup Y)$.

The dual feasibility is preserved because the corresponding rows of M satisfy the cut identity

$$M_X + M_Y = M_{X \cap Y} + M_{X \cup Y}.$$

Moreover, the objective value changes by

$$\varepsilon\big(f(X \cap Y) + f(X \cup Y) - f(X) - f(Y)\big) \leq 0$$

by crossing submodularity. Hence the objective does not increase. Repeating this uncrossing operation, and choosing an optimal solution with a suitable minimality condition, gives an optimal solution whose positive y-support is cross-free. □

The total unimodularity input

The remaining ingredient is a structural fact about cut-incidence matrices restricted to cross-free families.

Theorem 4.3. *Let $\mathcal{F} \subseteq 2^V$ be cross-free, and let $M_{\mathcal{F}}$ be the submatrix of M restricted to rows indexed by $\mathcal{F}$. Then $M_{\mathcal{F}}$ is totally unimodular.*

One way to prove this is to show that $M_{\mathcal{F}}$ is a network matrix. This is the content of the standard network-matrix argument; see, for instance, Schrijver, Theorem 13.21.

Combining the uncrossing claim with total unimodularity gives the desired integral optimal dual solution. Thus the submodular flow system is box-TDI.

Corollary 4.4 (Integrality of submodular flow polyhedra)**.** *Assume f is integer-valued and ℓ, r are integral. Then*

$$\{x \in \mathbb{R}^A : x(\delta^-(U)) - x(\delta^+(U)) \leq f(U)\ \forall U \in \mathcal{C},\ \ell \leq x \leq r\}$$

is an integral polyhedron.

5 Applications

5.1 Ordinary Circulations

The usual circulation condition in a digraph is

$$x(\delta^-(v)) = x(\delta^+(v)) \qquad \forall v \in V.$$

This can be encoded by submodular flow constraints by taking

$$\mathcal{C} = \{\{v\} : v \in V\}, \qquad f(\{v\}) = 0.$$

The inequalities are

$$x(\delta^-(v)) - x(\delta^+(v)) \leq 0 \qquad \forall v \in V.$$

Summing these inequalities over all vertices gives equality in total, so each individual inequality must in fact hold with equality. Therefore we recover ordinary circulation constraints.

5.2 Polymatroid Intersection

Let S be a finite ground set. A polymatroid intersection system has the form

$$x(U) \le f_1(U), \qquad x(U) \le f_2(U) \qquad \forall U \subseteq S,$$

together with $x \ge 0$, where f_1, f_2 are submodular functions on S. We previously proved that this system is box-TDI. We can reprove it by reducing it to submodular flows.

Let S_1 and S_2 be two copies of S, and let

$$V = S_1 \cup S_2.$$

For $U \subseteq S$, write $U_1 \subseteq S_1$ and $U_2 \subseteq S_2$ for the corresponding copies. Define

$$\mathcal{C} = \{U_1 : U \subseteq S\} \cup \{V \setminus U_2 : U \subseteq S\}.$$

This is a crossing family. Define $f : \mathcal{C} \to \mathbb{R}$ by

$$f(U_1) = f_1(U), \qquad f(V \setminus U_2) = f_2(U),$$

except on the set S_1, which appears in both descriptions. There we set

$$f(S_1) = \min\{f_1(S), f_2(S)\}.$$

A direct check shows that f is crossing submodular on $\mathcal{C}$.

Now define a digraph $D = (V, A)$ with one arc

$$s_2 \to s_1 \qquad \text{for each } s \in S.$$

We identify the variable on the arc $s_2 \to s_1$ with the variable $x(s)$.

For $Z = U_1$, we have

$$x(\delta^-(Z)) - x(\delta^+(Z)) = x(U),$$

so the submodular flow inequality gives

$$x(U) \le f_1(U).$$

For $Z = V \setminus U_2$, we again have

$$x(\delta^-(Z)) - x(\delta^+(Z)) = x(U),$$

and hence

$$x(U) \le f_2(U).$$

Together with the box constraint $x \ge 0$, this recovers the polymatroid intersection system. Therefore the box-TDI property follows from the Edmonds-Giles theorem.

5.3 Nash-Williams' Orientation Theorem

Theorem 5.1 (Nash-Williams)**.** *If an undirected graph G is $2k$-edge-connected, then it admits an orientation that is k-arc-connected.*

Proof sketch. Start from an arbitrary orientation $D = (V, A)$ of G. We want to reverse some arcs so that every nonempty proper set $U \subsetneq V$ has at least k arcs entering it.

Let $x(a) = 1$ if arc a is reversed, and $x(a) = 0$ otherwise. After reversing the arcs indicated by x, the number of arcs entering U becomes

$$|\delta^-(U)| - x(\delta^-(U)) + x(\delta^+(U)).$$

Thus the condition that at least k arcs enter every nonempty proper set U is equivalent to

$$x(\delta^-(U)) - x(\delta^+(U)) \le |\delta^-(U)| - k \qquad \forall \emptyset \neq U \subsetneq V.$$

This is a submodular flow system with

$$\mathcal{C} = \{U : \emptyset \neq U \subsetneq V\}, \qquad f(U) = |\delta^-(U)| - k,$$

together with the box constraints

$$0 \le x(a) \le 1 \qquad \forall a \in A.$$

The function $U \mapsto |\delta^-(U)|$ is crossing submodular, so the Edmonds-Giles theorem implies that the feasible region is integral whenever it is nonempty.

It remains only to prove nonemptiness. The vector $x(a) = 1/2$ for all $a \in A$ is feasible: indeed,

$$x(\delta^-(U)) - x(\delta^+(U)) = \frac{|\delta^-(U)| - |\delta^+(U)|}{2},$$

and the desired inequality is equivalent to

$$|\delta^-(U)| + |\delta^+(U)| \ge 2k,$$

which holds because G is $2k$-edge-connected.

Therefore the feasible region contains an integral point $x \in \{0, 1\}^A$. Reversing exactly the arcs with $x(a) = 1$ gives an orientation in which every nonempty proper set has at least k entering arcs. This is precisely k-arc-connectivity. □

5.4 The Lucchesi-Younger Theorem

Definition 5.2 (Directed cut)**.** A set of arcs $C \subseteq A$ is a *directed cut* if there exists a nonempty proper set $U \subsetneq V$ such that

$$C = \delta^-(U) \qquad \text{and} \qquad \delta^+(U) = \emptyset.$$

Remark 5.3. For a weakly connected digraph D, the absence of directed cuts is equivalent to strong connectivity.

Definition 5.4 (Dijoin)**.** A *dijoin* in $D = (V, A)$ is a set of arcs that intersects every directed cut of D.

Theorem 5.5 (Lucchesi-Younger)**.** *Let $D = (V, A)$ be a weakly connected digraph. Then the minimum size of a dijoin is equal to the maximum number of pairwise arc-disjoint directed cuts.*

Proof sketch. Let

$$\mathcal{C} = \{U : \emptyset \neq U \subsetneq V,\ \delta^+(U) = \emptyset\}.$$

Each $U \in \mathcal{C}$ corresponds to the directed cut $\delta^-(U)$. The family $\mathcal{C}$ is crossing, and the constant function

$$f(U) \equiv -1$$

is crossing submodular.

The Edmonds-Giles theorem applies to

$$z(\delta^-(U)) - z(\delta^+(U)) \leq -1 \qquad \forall U \in \mathcal{C},$$

with the box constraint $z \leq 0$. Since $\delta^+(U) = \emptyset$ for all $U \in \mathcal{C}$, this is

$$z(\delta^-(U)) \leq -1 \qquad \forall U \in \mathcal{C}, \qquad z \leq 0.$$

Setting $x = -z$, we obtain the dijoin covering system

$$x(\delta^-(U)) \geq 1 \qquad \forall U \in \mathcal{C}, \qquad x \geq 0.$$

Thus the minimum dijoin problem has the LP formulation

$$\begin{aligned} \min \quad & \sum_{a \in A} x(a) \\ \text{s.t.} \quad & x(\delta^-(U)) \geq 1 \quad \forall U \in \mathcal{C}, \\ & x(a) \geq 0 \qquad\quad \forall a \in A. \end{aligned}$$

By box-TDI and integrality of the right-hand side, this LP has an integral optimum, which is exactly a minimum-cardinality dijoin.

The dual LP is

$$\begin{aligned} \max \quad & \sum_{U \in \mathcal{C}} y(U) \\ \text{s.t.} \quad & \sum_{U : a \in \delta^-(U)} y(U) \leq 1 \quad \forall a \in A, \\ & y(U) \geq 0 \qquad\qquad\quad \forall U \in \mathcal{C}. \end{aligned}$$

Again, TDI implies that there is an integral optimal dual solution. Such a solution selects a collection of directed cuts, and the constraints say that no arc is used by more than one selected cut. Hence the dual optimum is the maximum number of pairwise arc-disjoint directed cuts.

By strong duality, the minimum size of a dijoin equals the maximum number of pairwise arc-disjoint directed cuts. □

Summary

The Edmonds-Giles theorem says that submodular flow systems are box-TDI. The proof combines two central ideas:

1. *Uncrossing:* crossing submodularity lets us replace a crossing dual support by a cross-free one without increasing the dual objective.

2. *Total unimodularity:* once the support is cross-free, the relevant cut-incidence matrix becomes a network matrix, hence totally unimodular.

This turns submodular flow systems into a unified integrality framework for several classical results, including polymatroid intersection, Nash-Williams' orientation theorem, and the Lucchesi-Younger dijoin theorem.